

Industry and Innovation Module

Business Model Canvas

**MediFlow Robotics: An Indoor Robotic
Delivery Service for Hospitals**

Group 4

Team members:

Yijie He 2617457

Shunan Guo 2617383

Zeyu Dong 2617661

Junru Duan 2617413

Yijie He 2617303

Yuang Zhou 2617376

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Note: Main work division is shown in the bracket

Other important work division includes:

Shunan Guo: Poster plot in Canva

Yuang Zhou: Responsible for the final report integration, revision and partial rewriting of all sections, redraw the pictures in PPT by using the materials and original pictures provided by group members except Figure CH-1 and Poster

Poster Demonstrations



MediFlow Robotics

An Indoor Robotic Delivery Service for Hospitals

SAFE, EFFICIENT AND RELIABLE INTRA-HOSPITAL LOGISTICS



中南大学 邓迪国际学院
Dundee International Institute of Central South University



Business Model Canvas Poster | Group 4

GROUP Members:
Yuang Zhou, Yijie He, Shunan Guo, Zeyu Dong, Junru Duan, Bowen Deng



THE SOLUTION

- Autonomous indoor robot delivery service
- Contactless and traceable transport
- Workflow integration with hospital routines
- Robot-as-a-Service business model

THE PROBLEM

- Repetitive manual delivery tasks consume staff time
- Cross-infection and handling risks
- Poor traceability in internal logistics
- Large hospitals require frequent multi-point deliveries

VALUE PROPOSITION



1.Reduce repetitive manual workload



2.Improve safety and hygiene



3. Provide reliable, traceable delivery



4.Support efficient hospital operations

Delivers medicines, lab samples, medical documents and routine supplies.

TARGET CUSTOMERS & USE SCENARIOS

Customer Segments






Key Use Scenarios






Why hospitals?
Hospital delivery requires safety, traceability, hygiene and workflow compatibility that general delivery solutions cannot easily provide.

KEY ACTIVITIES

Robot Design and Testing

Hardware Design, Sensor Integration, Robot Development, Safety Management, Redundant Testing

Software and Fleet Management

Navigation Software, Planning, Task Scheduling, Real-time Tracking, User Interface

Hospital Development and Workflow Integration

On-site Training, Remote Training, Hardware Training, Software Training, User Training

Maintenance and Safety Management

Hardware Maintenance, Software Updates, Safety Protocols, Emergency Response

MediFlow Robotics Delivery Service

Continuous Improvement

Feedback, Performance Analysis, Workflow Optimisation, Testing, Cost Reduction

KEY RESOURCES

PHYSICAL RESOURCES
Robots, charging stations, sensors, batteries

INTELLECTUAL RESOURCES
Software, map data, route optimisation, fleet management, cybersecurity

HUMAN EXPERTISE
Robotics engineers, software developers, maintenance technicians, workflow specialists

FINANCIAL CAPACITY
Investment, prototyping, manufacturing, pilot testing, maintenance funding

BUSINESS MODE SNAPSHOT

KEY PARTNERS

- AMR hardware suppliers
- Cloud providers
- Building / elevator integrators
- SPD partners
- Regulatory bodies

CHANNELS

- Direct hospital sales
- On-site deployment
- Training and support

CUSTOMER RELATIONSHIPS

- Dedicated support
- Staff training
- Long-term service continuity
- After-sales support

REVENUE STREAMS

- Monthly / annual subscriptions
- Implementation fees
- Usage-based charges
- Premium software licensing
- Maintenance and repair fees

COST STRUCTURE

- Hardware
- Software and platform
- Deployment
- Labour and support
- Maintenance
- Compliance and safety

EXPECTED IMPACT

- Save staff time for clinical work
- Reduce delivery errors and infection risk
- Improve logistics visibility and efficiency
- Offer a scalable and commercially viable hospital service

MediFlow Robotics combines engineering capability and commercial logic to deliver a practical, scalable hospital robot delivery service.

Note: Icons are from PPT, red notebook and figures are from Diligentrobots and author, the poster is plotted in Canva.

Introduction

In this Business Model Canvas research, we proposed MediFlow Robotics for indoor delivery service in hospital. The service is designed to support the movement of medicines, laboratory specimens and other clinical supplies inside hospitals. The idea is not to replace clinical staff, but to reduce avoidable delivery work and make internal logistics more reliable, traceable and easier to manage. In the following sections, more detail will be discussed for the business pattern for our proposed MediFlow Robotics.

Key Partners (Yijie He)

MediFlow Robotics requires a strategic partner ecosystem because hospital robotic delivery depends on several capabilities. The core business should focus on hospital workflow integration, management software and reliable delivery operation. Therefore, key partners are needed to provide special hardware, cloud infrastructure, building integration, hospital delivery access and healthcare compliance support. The most significant key partners include autonomous mobile robot (AMR) hardware suppliers; cloud infrastructure providers for internet of things (IoT) connection and remote monitoring; building integrators for elevator application programming interface (API) access; supply, processing and distribution (SPD) delivery partners and regulatory bodies such as the medicines and healthcare products regulatory agency (MHRA). The corresponding rationales are given in Table 1.

Table 1. Key partners and rationale for MediFlow Robotics

Partner category	What they provide	Rationale and evidence
Hardware and component suppliers	AMR base, sensors, batterie.	Reduces hardware burden; AMRs support hospital logistics tasks [1].
Cloud infrastructure providers	IoT connection, data storage, remote monitoring	Enables scalable fleet management; IoT Core manages device fleets [2].
Building and elevator integrators	Elevator API, door access, building-system integration	Reduces multi-floor deployment risk; Otis integrates service robots with elevators [3].
SPD delivery partners	Hospital supply-chain access	Supports local workflow integration; SPD models can

		improve hospital delivery efficiency [4].
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The logic behind these partnerships is practical rather. Hardware suppliers help MediFlow avoid heavy early investment in robot manufacturing, such as the suppliers of the chassis, sensors and batteries. Cloud partners make remote monitoring and data handling easier to scale, for instance, Amazon rent its cloud platform and service to the companies [2]. To safely move around in a hospital, robots also need to be able to get into the elevator, into access-controlled and restricted areas. For this reason, we need the support of building and elevator integrators. In terms of opening market access channels for SPD partners, they are very familiar with the processes of hospital supply departments, especially those of non-tertiary hospitals outside of large tertiary hospitals. Also, the support of regulatory and compliance partners can help to reduce uncertainties regarding safety, data processing and monitoring after the deployment of the robots.

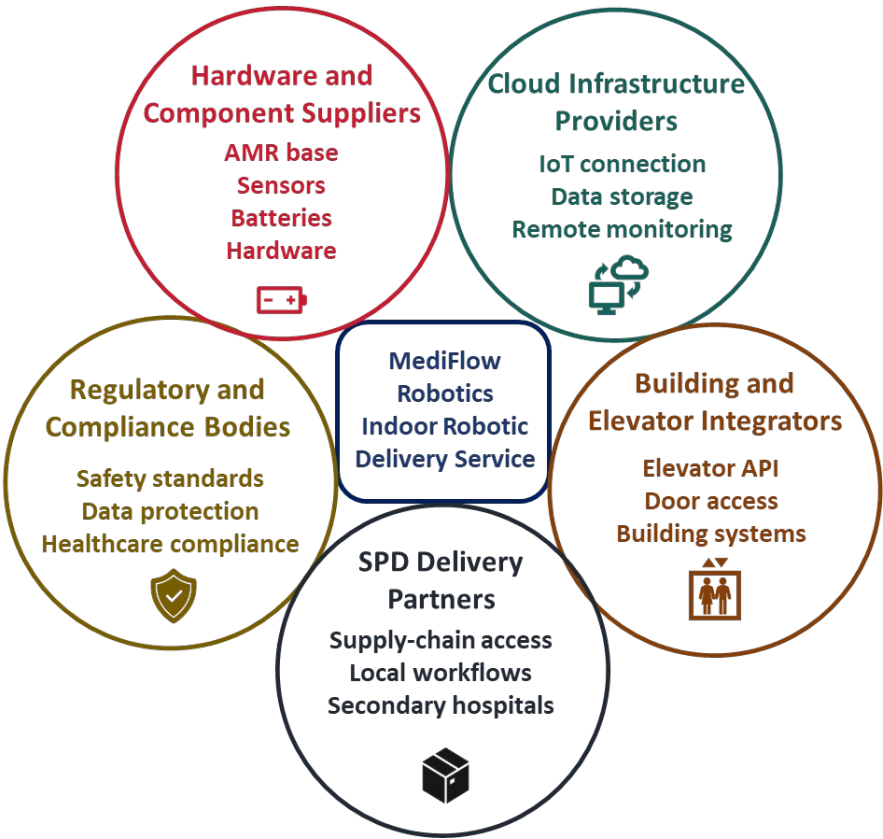


Figure 1. Key Partners ecosystem map in MediFlow Robotics (PPT plot).

Figure 1 displays the key partner ecosystem. The key partners of MediFlow Robotics do not replace the core of MediFlow Robotics. The hospital’s workflows, service offer and customer contact remain within MediFlow Robotics’ tasks and responsibilities. The key partners from various fields of activity provide important products and MediFlow Robotics’ medical procedures can be deployed safely and on a larger scale with the help of their infrastructure and knowledge. To understand the network of key partners from various fields of activity who provide key products for MediFlow Robotics in depth is important.

Key Activities (Shunan Guo)

For MediFlow Robotics, its key activities extend beyond the mere manufacturing of robots. The company must transform the robots into a service that can be tested, deployed, maintained, and improved within the actual hospital workflow. As shown in Figure 2, there are totally 5 parts: robot design and testing, software and cluster management, hospital deployment and process integration, maintenance and safety management, and continuous improvement.

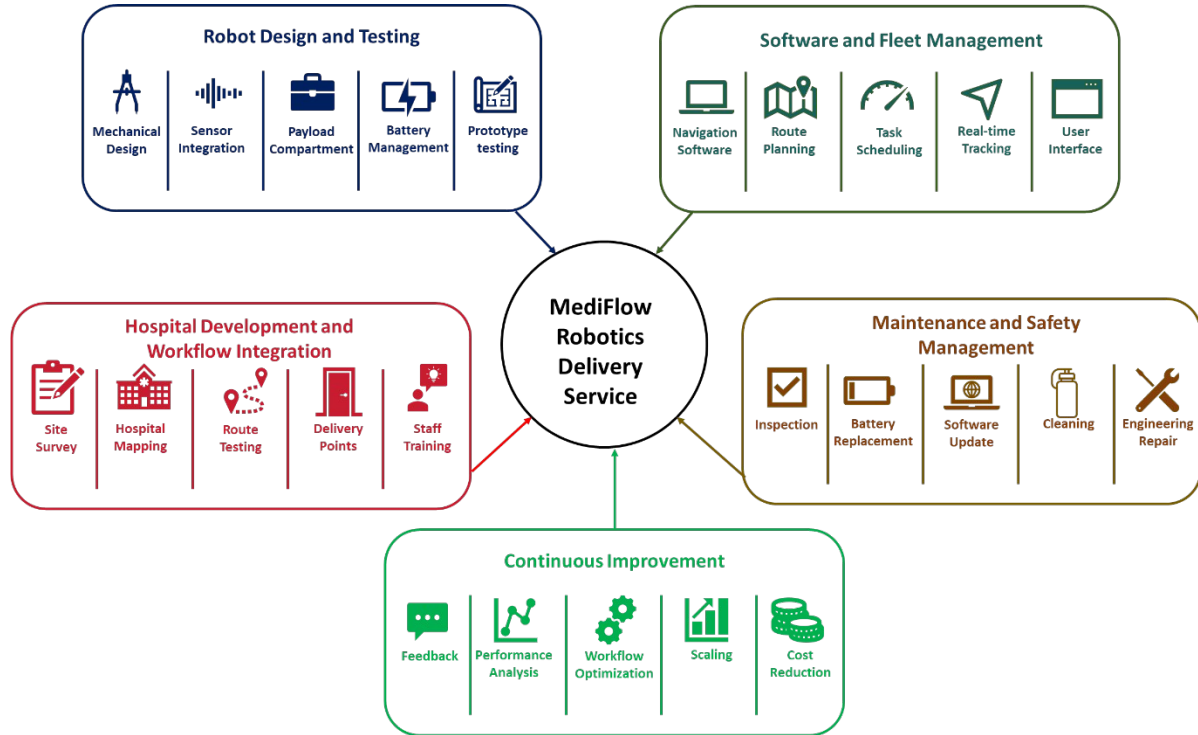


Figure 2. Key activities required to deliver in MediFlow Robotics (PPT plot).

Robot design and testing is a production activity. In MediFlow Robotics mechanical design, sensor integration, battery design and testing of prototypes are production activities which have to be done before a service can be launched in a hospital. To transport medicines and specimens through the hospital corridors safely, the robot must be more than just able to move around. Intelligent logistics robots can be used for drug and specimen delivery in hospitals, see also hospital robot research [6]. Safety testing must be done for securing the payload, for avoiding obstacles as well as for interaction with staff and patients. Different strategies for safety control of robots interacting with people exist and can be used as a general safety reference [7].

Software and fleet management: The platform behind MediFlow Robotics. Navigation software, route planning, task management, real-time tracking and user interface to manage multiple delivery jobs in different departments of a hospital. That is what hospitals are paying for with their subscription to a managed logistics service, including monitoring, updates and support.

Hospital deployment and workflow integration is where the main problem solving is taking place for MediFlow Robotics. Since each hospital has corridors, lifts, restricted areas and

specimen routes between departments which are vastly different from each other, it requires site surveys, hospital mapping, route testing and set up of delivery points as well as training of hospital staff to operate the system.

After the initial deployment of the service, maintenance and safety management continues to ensure the reliability of the service. This includes activities such as inspections, battery replacement, software updates, cleaning and engineering repair of robots, as well as data protection for delivery records and for access to hospital systems. As National Health Service (NHS) Digital states, organisations that hold confidential information about patients using their data must ensure the information is secure and is handled properly [8]. This is ensured through the feedback section, performance analysis of the current system, workflow optimisation and scaling up of service to meet demands.

Key Resources (Shunan Guo)

For MediFlow Robotics, the key resources are not just the robots themselves. Figure 3 separates the resources into four areas: physical resources, intellectual resources, human expertise and financial capacity.

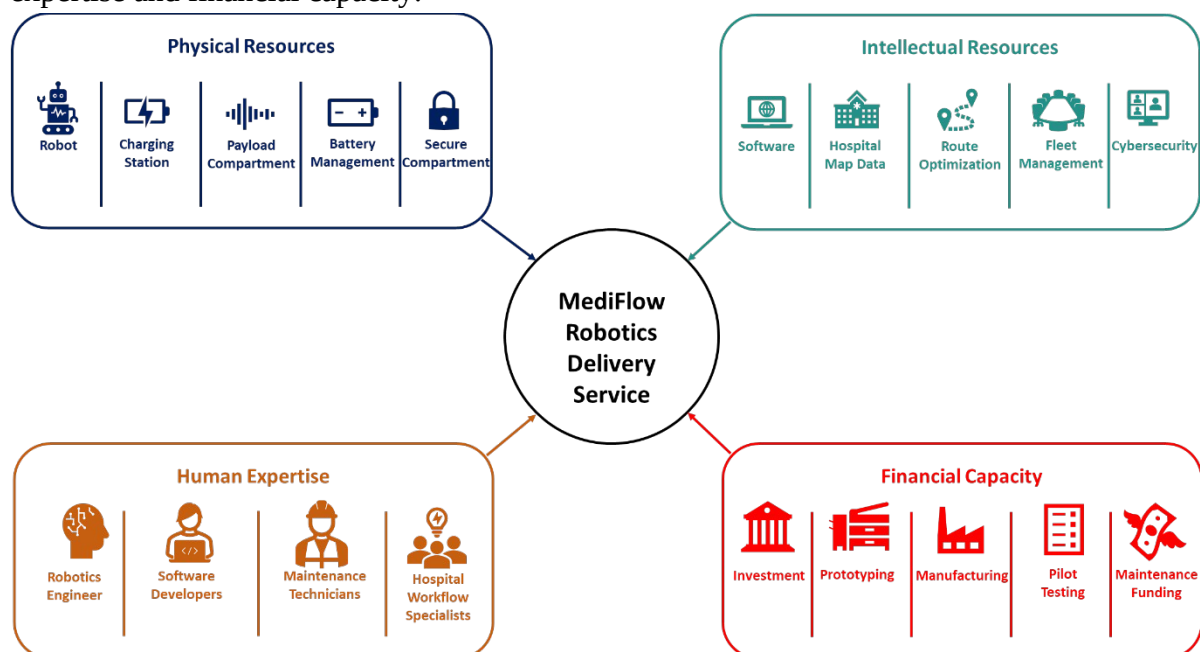


Figure 3. Key resources in MediFlow Robotics (PPT Plot).

From the perspective of the most visible resources, MediFlow Robotics require delivery robots, charging stations, payload containers, battery management and secure containers. These resources make hospital deliveries possible. A hospital robot has to move in corridors of the narrowest width, stop in any spot in the vicinity of staff and patients, carry medicines and specimens in safe containers.

The intellectual resources are difficult to collect. As shown in Figure 3, MediFlow Robotics includes navigation software, hospital map data, route optimisation, management knowledge and cybersecurity capability. A review of autonomous mobile robots in healthcare also shows

that healthcare robotics depends not only on the robot body, but also on the algorithms that make robots useful in hospital environments [9]. Cybersecurity knowledge is also important because the service may connect with hospital systems or delivery records. NHS Digital states that organisations with access to NHS patient data and systems are expected to present good data security [8].

Human expertise links these resources to real hospital use. Robotics engineers and software developers improve the robot and platform, while maintenance technicians keep the service running in good states. Hospital workflow specialists are also important since they understand how to manage different workflows (medicine, samples).

Financial capacity is the final block. MediFlow Robotics still need to get fund, manufacture, do pilot and maintain capabilities before subscription becomes stable. Overall, Figure 3 shows that MediFlow’s key resources form a complex system which allows the business to scale and become more stable.

Customer Relationships (Zeyu Dong)

For MediFlow Robotics, customer relationships should be significant in daily life. A hospital may only sign a contract, but the service will be used by lab teams and IT staff. If these groups have no ideas about who to contact or how to use the robots in their daily routines, the service will lose trust quickly. For this reason, MediFlow Robotics needs a relationship model built around clear responsibility, staff connection, remote operation and maintenance, and regular review. This fits the agency for healthcare research and quality (AHRQ) view that healthcare quality should include safety, timeliness and efficiency [10].

Table 2. Customer relationship types and rationale for MediFlow Robotics

Relationship type	How MediFlow delivers it	Rationale
Dedicated assistance	Account manager for hospital teams	Provides a clear contact and builds trust.
Training and onboarding	live training and refreshing sessions	Reduces resistance and errors.
Monitoring, operation and monitoring (O&M)	Remote dashboard, 24/7 support	Protects uptime and subscription value.
Continuous improvement	Usage reports and review meetings	Improves workflows and supports update.

There are several types of relationships that MediFlow Robotics is designed for. The first of these relationships is dedicated personal assistance. It is recommended that each hospital has an account manager who is responsible for Mediflow operations in collaboration with the

hospital's pharmacy and laboratory departments. The role of dedicated personal assistance is to facilitate communication between clinical users, the delivery teams and the technical support staff. To this end, a single contact point for daily communication and operations is created. Moreover, hospital staff must be trained and assisted with setting up the service. After that, the service can be adjusted to the user's routines, and he or she can respond to alerts that the robot sends out.

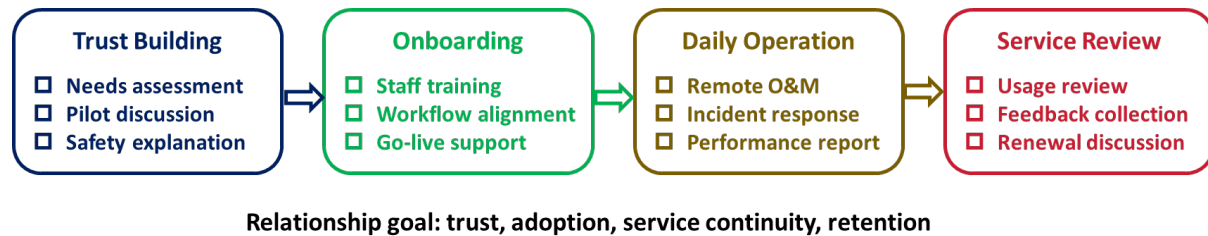


Figure 4. Customer relationship lifecycle in MediFlow Robotics (PPT plot, 2026).

Figure 4 presents the relationship as a lifecycle. The early stage is about trust building; after that, onboarding focuses on staff training and workflow alignment. After the service is launched, the relationship focuses on the service's operation and monitoring, incident handling and performance measurement. For example, Aethon's hospital robot solution includes 24/7 connected support and service maintenance after the solution has been launched [11].

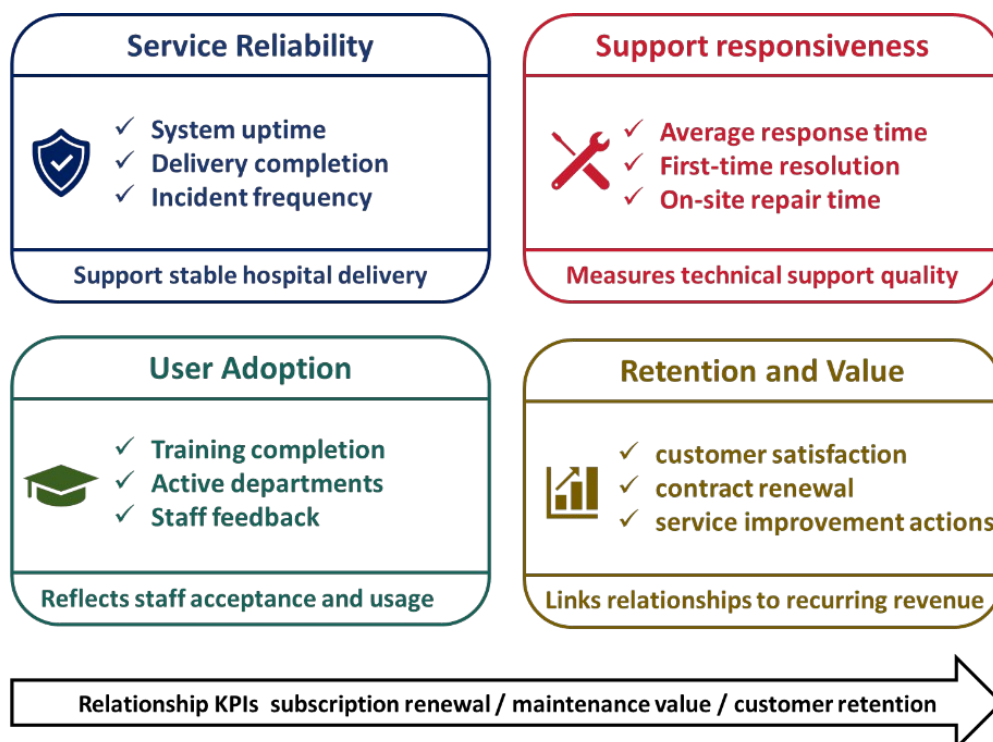


Figure 5. Relationship key performance indicators (KPIs) in MediFlow Robotics (PPT plot, 2026).

Figure 5 adds measures to the relationships model. By introducing Key Performance Indicators (KPIs) such as system availability, reaction time, first time fix and customer satisfaction, MediFlow Robotics can start to measure and govern the relationship to determine whether it is working or not. The Institute for healthcare improvement (IHI) states

that in order to deliver safe and reliable health care, healthcare organizations need to support learning, and improvement and measure their work [12].

Channels (Junru Duan)

Hospitals will be attracted by MediFlow Robotics through a mix of direct and indirect sales channels. On the direct side, hospitals of Grade III Level A (i.e., Chinese Tier 1 hospitals) are more complex in terms of their logistics, have larger purchasing budgets and require greater safety and reliability, and so are best sold to by an internal team of account managers, technical specialists and deployment managers. Indirect sales will be through supply, processing and distribution (SPD) providers, who are familiar with the hospital's supply-chain workflow and can help the company attract even more customers [13][14].

As shown in Figure 6, the channel strategy follows the five channel phases: Awareness, evaluation, purchase, delivery and after-sales support. In the awareness stage, MediFlow Robotics will use indirect channels such as academic platforms, professional networks and SPD partner endorsement to build credibility. This is important because hospitals want to reduce risk when adopting clinical logistics technology. Third-party validation can reduce perceived adoption risk before hospitals move into direct evaluation. Third-party validation can reduce perceived adoption risk before hospitals move into direct evaluation.

	● Direct Channel	● Indirect Channel
Awareness	Third-party Validation like Academic Platform	SPD Partnership Endorsement
Evaluation	Third-party Validation like Academic Platform	Handover to direct channel
Purchase	30-day Free Trial	-
Delivery	Serving in Hospital	-
After-sale	24/7 remote O&M Rapid 4-hour response	-

Figure 6. Channel Strategy Overview (Canva Plot, 2026)

During evaluation, MediFlow Robotics can use performance evidence to support the discussion, Li et al. reported that robot delivery reduced delivery time by 32% to 36% compared with manual delivery in a tertiary hospital study [6]. This provides useful evidence, but the purpose of a pilot is to test the local situation: routes, lift access, delivery points and task volume.

The purchase stages require direct control. Public hospital procurement often involves technical clarification and compliance checks. China's Government Procurement Law defines government procurement as purchasing activities conducted with funds by government departments and public organisations, so a technical team should handle tender documents and procurement communication [15]. After purchase, direct service is still needed for on-site installation, Hospital information system (HIS) integration and staff training. SPD partners can help with local workflow, but the technical deployment should remain under control. After-sales support should also stay mainly direct. MediFlow Robotics should provide remote operation and maintenance, monitoring and rapid on-site response when required.

Customer Segments (Bowen Deng)

For our MediFlow Robots service designed for hospitals, we have carried out clear and detailed customer segmentation. The core questions are: What kind of market model should be adopted to classify customers? For whom are we creating value? These questions will be answered in the subsequent paragraphs

For mass market, it is not applicable to MediFlow Robot. Delivery services for the mass market (such as food delivery and express delivery) serve highly homogeneous demands. However, hospital scenarios impose strict requirements on delivery safety, traceability, disinfection compliance, and path planning, which cannot be met by general delivery solutions. Therefore, we do not target the mass market.

For niche market, this is one of the core positioning strategies of MediFlow Robotics. We focus on the highly vertical segment of intra-city delivery for hospital scenarios. Our target customers are medical institutions including public hospitals, private hospitals, and community health centres. Their demand is highly concentrated on safe, efficient, and low-infection-risk delivery of various medical supplies, which clearly falls into the niche market category.

For segmented market, this is our refined strategy within the niche market. We further divide our core hospital customers into differentiated groups and provide targeted services according to their specific needs.

For diversified market, this model is not adopted at the current stage but is reserved for future expansion. In the long run, robot technology can be extended to diversified markets, such as material delivery services for nursing homes, rehabilitation centres, and pharmaceutical enterprises. These scenarios differ greatly from hospitals and will be explored as future business directions.

For multi-sided platform, this is not a core model at present but can be developed as a long-term ecological plan. Currently, our direct customers are medical institutions. In the future, we plan to build a multi-sided platform that serves two or more interdependent user groups simultaneously. The market type analysis is summarised in Table 3.

Table 3. Market type analysis in MediFlow Robots

Market Type	Applicability	Core Description
Mass Market	No	Hospital delivery has stricter professional requirements
Niche Market	Core point	Focus on hospital scenarios and provide medical material delivery.
Segmented Market	Main strategy	Provide services within hospitals based on scale, departments, and supply types.
Diversified Market	Future option	Expand to nursing homes and pharmaceutical enterprises.
Multi-sided Platform	Long-term plan	Connect hospitals and testing institutions to form an integrated platform.

From different perspectives, customers can be divided into different groups with distinct pain points and demands. MediFlow Robotics use these perspectives to do the customer segmentation. These can be summarised in Table 4 and 5.

Table 4. Segmentation by job role

Customer Role	Typical Users	Core Pain Points	Core Needs
Decision-makers	Hospital directors	High operating costs for material delivery.	Reduce long-term labor costs and traceable delivery procedures.
Frontline Staff	Nurses, laboratory technicians	Delivery tasks disrupt clinical work	Reduce manual labor and unnecessary movement and avoid waste.
Logistics Department	Delivery staff	Heavy manual delivery workload and low efficiency	Automated delivery; reduce reliance on manual labor.

Table 5. Segmentation by hospital scale and type

Customer Type	Typical Features	Core Needs
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Large Hospitals	Many departments and complex workflows	Multi-robot scheduling, cross-building delivery.
Small Hospitals	Limited space, low budget, stable delivery workload	Low cost, easy deployment, simple operation
Specialized Hospitals	Maternity, infectious disease hospitals	Low volume but high-standard delivery; strict control over temperature for safety.
Third-party Medical Institutions	Independent testing centers, blood stations	Batch transportation of samples; full-process data traceability.

In conclusion, we have analysed market type and core customer demands and completed multi-dimensional customer segmentation. Based on these findings, MediFlow Robotics will create value for target customers and solve their practical pain points.

Revenue Streams (Yuang Zhou)

MediFlow Robotics will adopt a hybrid revenue structure, including monthly or annual robot service subscriptions, one-off implementation fees, usage-based charges, premium software licensing and hardware upgrade, maintenance and repair fees. The total revenue structure is summarised in Table 6, and the revenue streams with rationales will be discussed in detail in subsequent paragraphs.

Table 6. Hybrid revenue model and rationale for MediFlow Robotics

Revenue streams	What the hospital pays for	Payment basis	Rationale
Monthly or annual robot service subscriptions	Robot access, remote monitoring, software updates, technical support	Per robot, paid monthly or annually	Relay Robotics states that its robots are available through monthly subscription Error: Reference source not found.
One-off implementation fees	Hospital mapping, workflow analysis, charging dock installation, staff training	One-time fee at the start of deployment	Aethon's hospital robot solution includes technical integration before official launch[11].
Usage-based	Extra delivery tasks,	Per task or per	Rochester Regional

charges	extra operating hours. high-volume use beyond the basic subscription	operating hour	Health pays for Moxi based on how much the robots work Error: Reference source not found.
Premium software licensing	Temperature monitoring, hospital system integration, analytics dashboard	Annual licence or add-on module fee	NHS states that medicine temperature should be managed in delivery Error: Reference source not found.
Hardware upgrade, maintenance and repair fees	Battery replacement, temperature-controlled modules and out-of-contract repairs	Annual maintenance contract or pay-per-repair	Aethon presents hospital robots as a complete solution with service support [11].

The rationale for this hybrid model can be concluded as: reliable, traceable and continuously supported service. Diligent Robotics describes Moxi as supporting hospitals with tasks such as running patient supplies, delivering lab samples, fetching central supply items and delivering medications[19], with product figure showing in Figure 7. This shows that the customer value lies in reducing repetitive tasks and improving reliable long-term service.

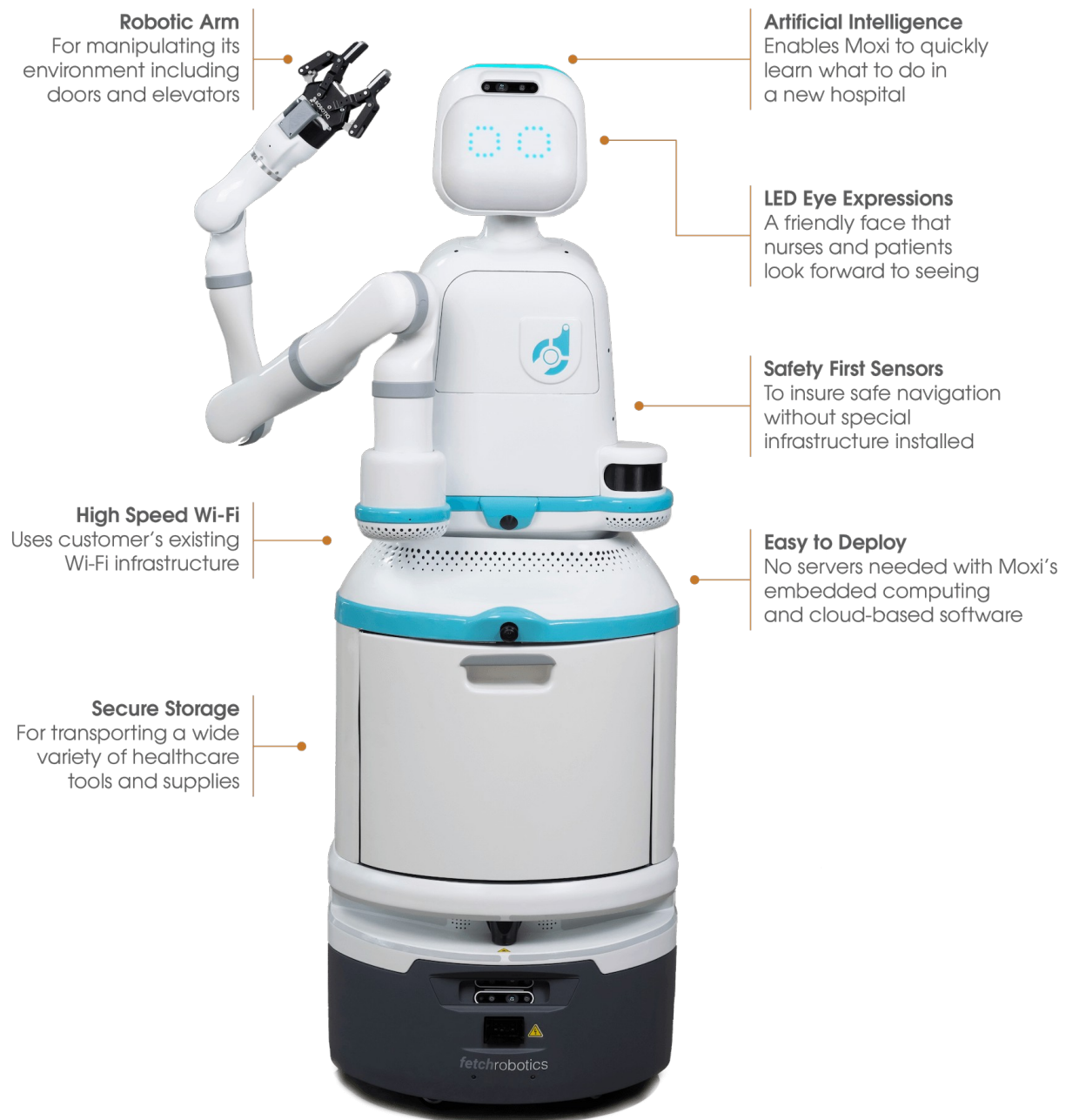


Figure 7. Moxi hospital service robot (Diligentrobots, 2026). Available at:
<https://www.diligentrobots.com/moxi>

Furthermore, Figure 8 supports this value-based pricing rationale. As can be seen in the graph, the workforce of NHS (National Health Service) is continuously increasing over time, while the annual leaver rate remains around 10 %, which indicates that more automation services could be adopted to reduce avoidable repetitive workload. It does not suggest that robots should replace staff. Instead, it supports the pricing of the robots as a service that improves the efficiency and assist the clinical staff to focus more on their own work.

NHS workforce scale and turnover pressure as a background for hospital logistics automation

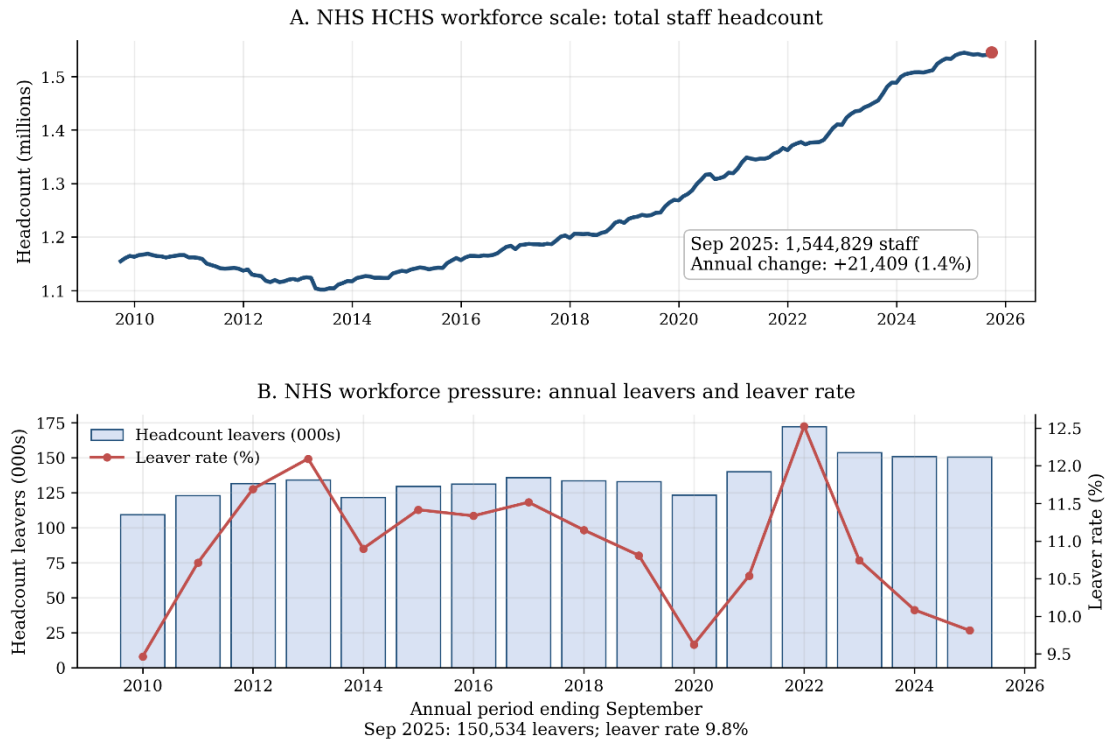


Figure 8. NHS workforce scale and turnover pressure as background evidence for hospital logistics automation (Python plot, 2026). Available at:

<https://www.nhsemployers.org/articles/pay-scales-202526>, <https://digital.nhs.uk/data-and-information/publications/statistical/nhs-workforce-statistics/september-2025>

Based on this rationale, the main revenue stream should be the monthly or annual robot service subscription, because it provides stable income while reducing the hospital's upfront purchasing burden. Market examples support this service-based model: Relay Robotics offers hospital delivery robots through subscription, and Rochester Regional Health pays for Moxi based on how much the robots work [16][17]. The structure of subscription should be the core service package, including robot access, remote monitoring, software updates, technical support and standard maintenance. Other revenue streams, including implementation fees, usage-based charges, premium software licensing, and hardware, should cover additional value not included in the basic subscription, such as hospital mapping, staff training and out-of-contract repairs [11].

In contrast, some pricing patterns such as advertising and commission are not that suitable to be used in MediFlow Robotics. Hospital deliveries involve medicines, laboratory samples and dangerous chemical reagents, which should not depend on advertising or other commission. A free or discounted plan might be employed as a short-term strategy in order to get a long-term cooperation. Overall, the revenue model should be value-driven: the hospital pays for reliable delivery performance, deployment support, traceability and long-term service continuity. To address how each revenue stream contributes to overall revenues, Figure 9 presents an estimated revenue mix. The percentages are author-created assumptions based on the proposed business model rather than market-wide statistics.

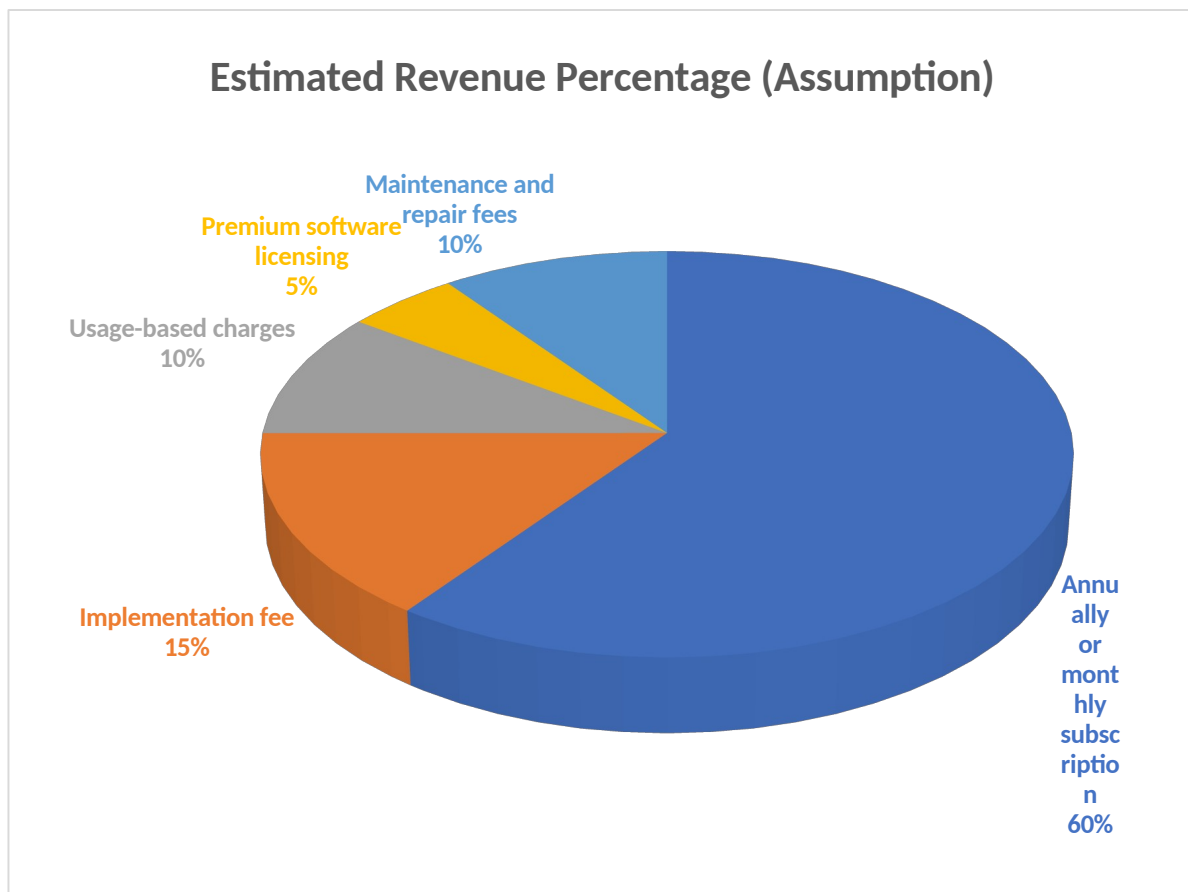


Figure 9. Estimated revenue mix for MediFlow Robotics. (Excel plot, 2026)

Cost Structure (Yuang Zhou)

MediFlow Robotics will adopt a value-driven cost structure rather. This is because hospitals do not only require a low-cost delivery robot, what they truly require is safe, reliable, traceable and continuously service. Therefore, the main costs should not be limited to the robot hardware itself; they are supposed to expand to the wider service system which requires deployment, integration, operation and support as reliable hospital delivery service.

Cost Structure Map



Figure 10. Cost structure map in MediFlow Robotics (Visio plot, 2026). Icons are generated in Flaticon <https://www.flaticon.com/free-icons/generator>

Figure 10 summarises the main cost structure in MediFlow Robotics. Hardware costs are linked to the robot chassis, sensors, batteries, payload modules and charging docks. These are variable costs because they increase as more robots are deployed [16]. Autonomous delivery also depends on specific sensors such as cameras and lidars. Software and platform costs are also significant: navigation software, remote monitoring and the management system require substantial development before the service can be scaled, but the same platform can be used across more hospital customers after that.

Table 7. Rationales with cost categories for MediFlow Robotics

Cost categories	Main cost items	Fixed or Variable	Rationale
Hardware costs	Chassis, sensors, batteries, payload	Mostly variable	Safe autonomous delivery requires

	modules, charging docks		sensing hardware such as cameras and lidars [16].
Software and platform costs	Navigation software, management system, remote monitoring, updates	Mostly fixed and partly variable	Specific software is needed to manage robot operations and measure service output [11].
Deployment costs	Mapping, dock installation, staff training, elevator and door integration	Variable per hospital	Hospital robots require deployment, including elevators and building systems [11]
Operation and maintenance costs	Remote support, repairs, spare parts, battery replacement	Variable	Continuous support and maintenance are required to keep robots reliable in operation [11].
Labour and support costs	Engineers, field service staff, project management	Fixed and variable	Skilled staff are needed for deployment and ongoing technical support [11].
Compliance and safety costs	Risk assessment, data protection, safety testing, insurance	Mostly fixed and partly variable	Hospital delivery involves medicines and sensitive data [8][18].

Table 7 summarises the major cost categories and corresponding rationales. Deployment costs are especially significant because every hospital has its own lifts, security rules and delivery routines. MediFlow Robotics would need mapping, staff training and building integration before operation. Operation and maintenance costs also remain vital after deployment, since hospitals expect continuous service. Batteries, spare parts, repairs and remote support will keep the service reliable in long-term operation.

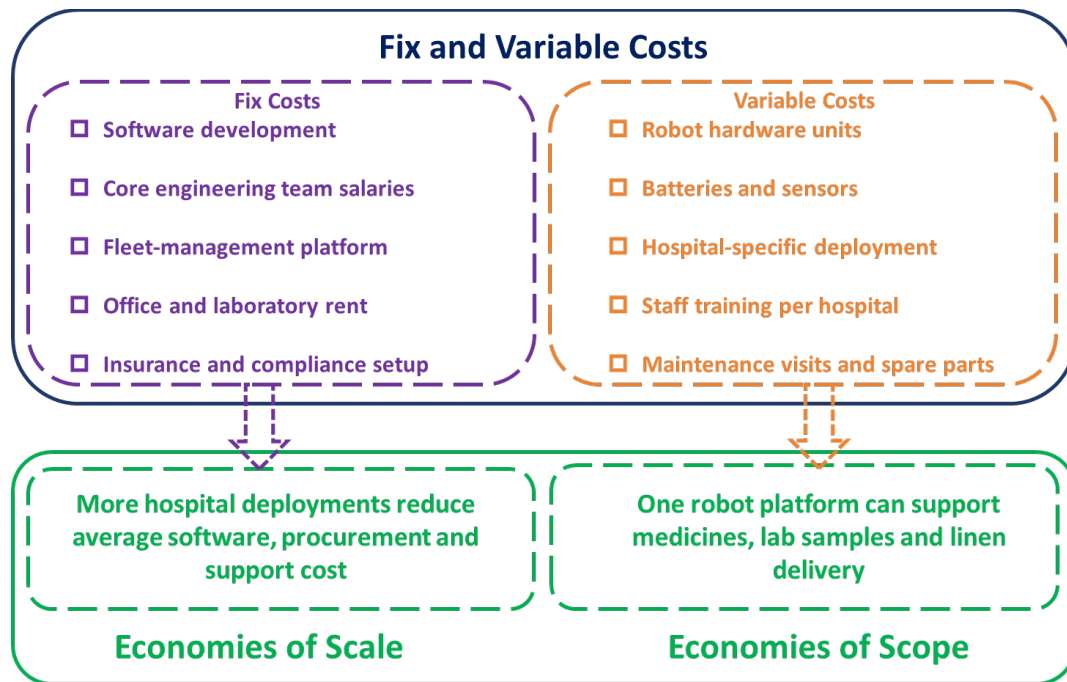


Figure 11. Fixed and variable cost structure in MediFlow Robotics (PPT plot, 2026).

Figure 11 outlines fixed and variable costs for MediFlow Robotics. Software development, management platform, compliance setup, core engineering salaries are examples of fixed costs that increase as sales increase. Variable costs include robot hardware, deployment, staff training, and spare parts. As more hospitals are using MediFlow Robotics platform costs will become cheaper as they can be distributed over a larger customer base creating economies of scale. Also, economies of scope are created as the same robot platform can be used for different tasks, like for example: medicines, test samples and other medical materials delivery. [18].

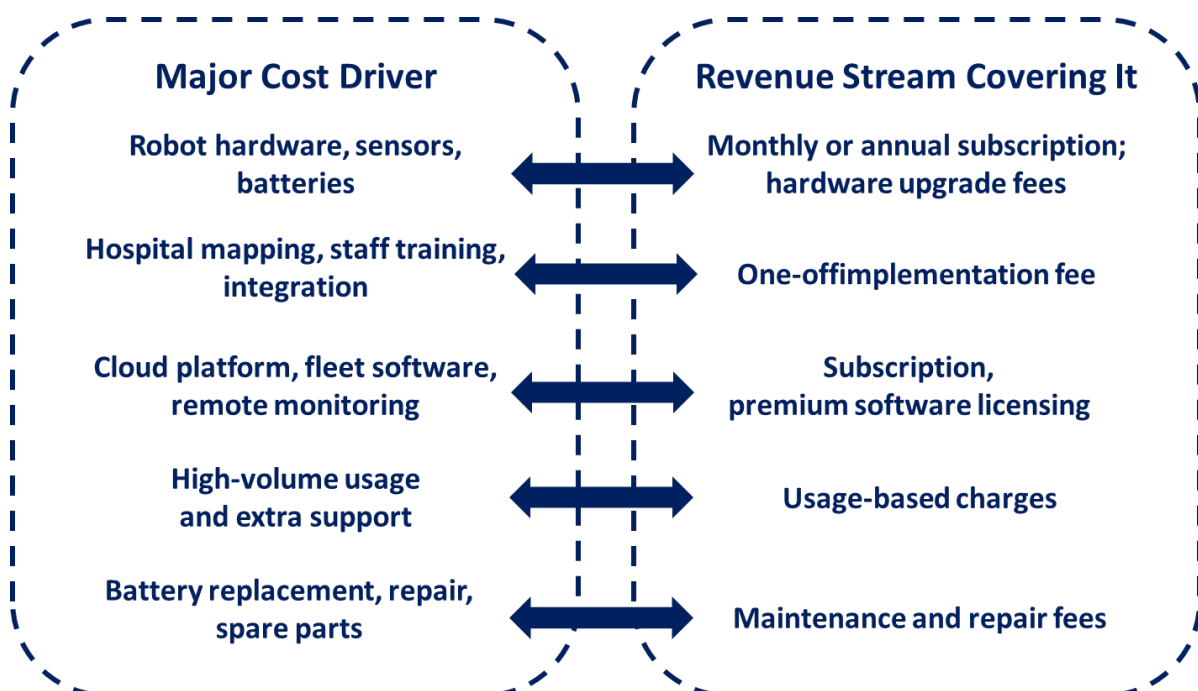


Figure 12. Cost–Revenue map in MediFlow Robotics (PPT plot, 2026).

Figure 12 highlights relationship between cost structure and revenue streams. The main subscription fee is used to cover the cost of the robots as well as the cost of remote monitoring, updates and standard maintenance for the robots. The upfront fee for implementation is used to pay for the initial mapping as well as the installation, training and technical integration. The usage-based charge is to cover increased volume of deliveries for customers. Lastly, the maintenance and repair fees cover the cost of batteries as well as spare parts for under warranty repairs as well as out-of-contract repairs.

Value Proposition (Group 4)

MediFlow Robotics starts from a practical problem in hospitals: internal delivery is significant, but lots of packages and materials are repetitive, time-consuming and difficult to be delivered. Medicines, specimens and other clinical supplies must move between pharmacies, laboratories and storage areas every day. When this work only depends on clinical staffs, delays and interruptions can occur and enormously affect operation efficiency. Thus, MediFlow Robotics proposes its own value proposition: removing avoidable transport work from clinical worker's daily job and make internal deliveries more reliable.

The service model offered by MediFlow Robotics is a mixture of hardware, software and bonus support. The physical robot provides autonomous indoor delivery and safe transport. The software provides mapping, route planning, and task handling. The service adds installation, staff training, remote support, maintenance and continuous improvement. This combination matters because hospitals are not simply buying a real robot. They need a sustainable and specific service that can fit into their own hospital corridors, lift and delivery routes.

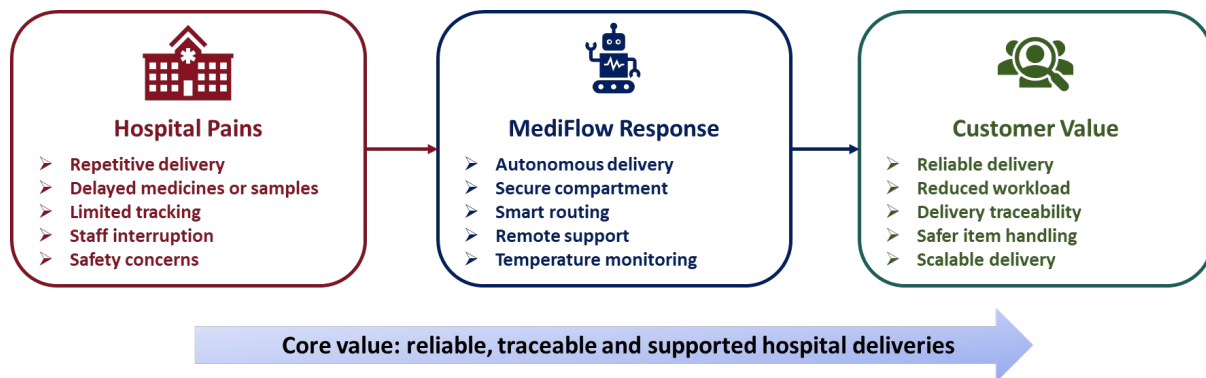


Figure 13. Value proposition logic diagram in MediFlow Robotics (PPT plot, 2026)

The main customer value is reliability. Hospital delivery tasks often involve items that cannot easily arrive; medicines and specimens may be sensitive to time, and staff need confidence that a delivery has been completed. MediFlow Robotics cannot assume that every hospital will achieve the same result, but this evidence supports the idea that robotic delivery can improve internal transport performance when it is properly implemented.

A second value is risk reduction. Hospital delivery is not the same as ordinary parcel delivery. Some medicines require controlled temperature, or other conditions, some samples

need to be traced, and hospital staff need clear records of where items have gone. NHS specialist pharmacy service notes that transporting medicines requires attention to suitable packaging, labelling, audit trail and temperature monitoring for cold-chain medicines [18]. MediFlow Robotics can support this need through secure compartments, delivery records and optional temperature-monitoring modules. This makes the service useful not only for routine consumables, but also for higher-sensitivity delivery tasks.

A third value is better usage of staff time. The NHS workforce data show the scale of hospital labour systems: NHS hospital and community health services (HCHS) staff headcount reached 1,544,829 in September 2025 [20]. Large hospital workforces create many internal deliveries flows, and even small, repeated transport tasks can add up across departments. MediFlow Robotics helps by taking over routine movement tasks, so nurses, pharmacy staff and delivery teams can focus more on work that requires human judgement or direct patient support.

The value proposition is also customised for different stakeholders. Hospital operations managers gain better visibility and standardisation of internal delivery. Pharmacy and laboratory teams gain traceable and more predictable movement of medicines and specimens. Clinical staff benefit from fewer interruptions caused by non-clinical transport work. IT benefits from a service model that includes mapping, integration and maintenance rather than leaving the hospital to solve technical issues alone.

Overall, MediFlow Robotics delivers value through performance, risk reduction, convenience and “getting the job done”. Its value is not based on novelty alone. The stronger argument is that hospitals need internal logistics that are safe, traceable, supported and adaptable to local workflows. By combining autonomous delivery robots with software, technical support and service continuity, MediFlow Robotics offers a practical value proposition for hospitals with high internal delivery demand.

Conclusion

To sum up our proposed MediFlow Robotics, the research shows that MediFlow Robotics can create value by reducing repetitive delivery tasks, improving visibility over hospital delivery and supporting staff to focus more on clinical or operational work. The model is commercially realistic because its core value presents long-term reliability, traceability and support. Therefore, MediFlow Robotics is promising pattern as a hospital logistics service, not only as an indoor delivery robot.

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